

PENN VALLEY, PA, USA

WMO: 725105

Lat: 40.821N

Lon: 76.864W

Elev: 135

StdP: 99.71

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14770

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-12.0	-21.0	0.6	-13.0	-18.3	0.8	-9.9	9.1	-2.4	8.2	-2.5	1.7	340	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	32.6	23.4	31.2	22.7	29.7	22.0	25.0	30.1	24.1	28.8	23.3	27.8	2.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.5	27.3	22.6	17.6	26.5	22.0	17.0	25.9	76.3	30.1	72.9	28.7	69.9	27.7	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.9	5.6	-18.9	35.1	3.3	1.9	-21.3	36.5	-23.2	37.6	-25.1	38.7	-27.5	40.1		
(5)			-19.3	26.8	3.2	1.1	-21.5	27.6	-23.4	28.3	-25.2	28.9	-27.5	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-2.2	-1.0	4.1	10.5	16.3	21.1	23.0	22.0	18.2	12.0	5.8	0.7
	DBStd	9.87	5.58	5.05	5.25	4.72	4.29	3.30	2.81	2.71	3.68	4.50	4.42	4.70
	HDD10.0	1394	379	308	196	49	3	0	0	0	0	30	139	291
	HDD18.3	3144	637	541	442	238	93	13	2	4	48	202	377	548
	CDD10.0	1734	1	1	12	64	198	332	403	371	246	92	13	2
	CDD18.3	441	0	0	0	3	31	94	147	116	43	6	0	0
	CDH23.3	3988	0	0	4	86	325	852	1327	959	384	51	0	0
	CDH26.7	1304	0	0	0	24	97	274	489	301	109	9	0	0
(14) Wind	WSAvg	1.9	2.3	2.4	2.7	2.6	2.0	1.8	1.3	1.1	1.3	1.6	2.0	2.1
(15) Precipitation	PrecAvg	1039	73	61	79	86	94	112	95	94	104	81	86	76
	PrecMax	1535	199	178	160	224	221	460	182	237	305	223	190	178
	PrecMin	753	11	8	21	20	10	10	32	14	19	2	21	10
	PrecStd	185	46	36	33	43	43	71	42	48	65	51	38	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	17.2	23.0	30.0	32.0	33.5	35.2	33.9	32.1	27.5	21.1	16.8
		MCWB	13.1	12.4	14.0	17.9	21.5	23.5	24.2	23.7	22.6	20.6	15.4	13.5
	2%	DB	11.3	12.0	18.9	25.9	29.4	31.5	32.7	31.5	29.7	24.3	18.0	12.4
		MCWB	8.9	7.7	11.8	16.3	20.3	22.4	24.0	23.6	21.8	18.4	13.7	9.6
	5%	DB	7.6	8.8	15.6	22.5	27.1	29.7	31.1	29.7	27.5	22.2	16.0	9.5
		MCWB	5.0	5.3	10.3	14.3	19.2	21.8	23.2	22.6	20.6	17.4	12.0	7.4
	10%	DB	5.5	6.3	12.6	19.4	24.6	27.9	29.4	28.1	25.5	19.9	13.6	7.5
		MCWB	3.3	3.4	8.4	12.5	17.9	20.9	22.4	21.8	19.6	15.8	10.2	5.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.7	16.0	20.2	23.4	24.8	26.3	26.1	24.7	22.0	17.0	14.5
		MCDB	15.6	16.7	20.8	26.2	28.7	30.5	32.2	30.9	29.4	25.7	19.2	16.3
	2%	WB	8.9	8.5	13.6	17.5	21.7	23.9	25.1	24.8	23.1	20.4	14.9	10.4
		MCDB	10.8	11.9	17.5	23.3	27.4	29.3	30.6	29.2	27.4	23.5	17.5	11.8
	5%	WB	5.6	5.8	11.2	15.7	20.5	23.0	24.2	23.8	22.0	18.4	12.7	7.7
		MCDB	7.3	8.1	15.0	21.0	25.3	27.9	29.0	27.9	25.4	21.1	15.0	9.3
	10%	WB	3.5	3.6	8.8	13.8	19.1	22.1	23.4	22.9	20.9	16.3	10.7	5.6
		MCDB	5.2	6.1	12.2	18.2	23.2	26.4	27.9	26.7	23.8	18.8	13.1	7.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.7	10.9	12.6	12.3	11.5	12.0	11.8	12.5	11.6	10.5	8.0
		MCDBR	11.1	13.8	15.9	18.2	15.5	13.3	13.5	13.5	14.9	14.7	14.3	10.3
		MCWBR	8.9	10.0	9.9	9.5	7.6	5.9	5.9	6.5	7.6	8.6	9.8	8.5
	5% WB	MCDBR	10.4	12.3	14.1	15.3	13.0	11.6	11.5	11.3	11.8	12.0	11.6	9.3
		MCWBR	8.9	9.7	9.8	9.2	7.0	5.7	5.5	6.0	6.7	7.8	9.1	8.4
(40) Clear-Sky Solar Irradiance	taub		0.316	0.338	0.372	0.390	0.423	0.489	0.494	0.489	0.405	0.375	0.332	0.313
	taud		2.394	2.310	2.301	2.297	2.234	2.084	2.083	2.083	2.294	2.381	2.465	2.449
	Ebn at Noon		835	867	875	882	857	798	789	780	830	816	813	811
	Edh at Noon		83	104	118	126	138	161	160	155	117	97	77	73
(44) All-Sky Solar Radiation	RadAvg		1.71	2.60	3.54	4.59	5.20	5.71	5.80	5.08	4.14	2.85	1.97	1.42
	RadStd		0.14	0.32	0.28	0.47	0.58	0.49	0.35	0.29	0.54	0.33	0.22	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+56

Nomenclature: See separate page